

DAVID THAI

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EDUCATION

University of California, Berkeley

August 2021 – May 2025

Bachelor of Arts in Computer Science, Bachelor of Arts in Data Science

Berkeley, CA

- Relevant Coursework: Data Structures, Efficient Algorithms, Software Engineering, Computer Security, Database Systems, Artificial Intelligence, Computer Architecture, Data Mining & Analytics, Machine Learning

TECHNICAL SKILLS

Languages/Frameworks: Python (Flask), Java, Go, C++, JavaScript (React, Redux, Node.js, Express, jQuery), C, SQL (PostgreSQL, MySQL), HTML, CSS (Bootstrap, Tailwind), Ruby on Rails, MongoDB, Socket.io

Tools: Git, VS Code, IntelliJ, Heroku, Netlify, JUnit, Cucumber, Capybara, RSpec, Jupyter, Docker, Kafka, Prometheus, Grafana

Libraries: pandas, NumPy, Matplotlib, scikit-learn, seaborn, SciPy

PROFESSIONAL EXPERIENCE

Software Developer

May 2023 – August 2023

University of California, Berkeley — ActionMap Project

Berkeley, CA

- Led development and maintenance of a legacy civic info platform in Ruby on Rails and jQuery, providing information for 63 states with 113 geographic data files.
- Reduced page load time by 38% by refactoring UI components and optimizing 19 JavaScript dependencies.
- Led a 5-engineer team as Scrum Master, enforcing Agile practices via sprint planning, standups, and retrospectives.
- Improved automated test coverage to 56.7% using RSpec (14 files) and Cucumber (3 files) with 12 test methods and 9 scenarios, and Capybara in Test-Driven and Behavior-Driven Development.

Computer Science Mentor / Finance Committee Officer

Jan 2024 – May 2025

Berkeley ANova

Berkeley, CA

- Taught introductory CS concepts and programming (HTML, CSS, JavaScript, Python, pandas) to 20+ under-resourced Bay Area students per semester.
- Raised over \$4,000 by leading fundraising efforts in 2025 and managing organizational finances.

PROJECTS

ReChat: Real-time Distributed Chat Platform | React, Node.js, Express, MongoDB, Socket.io, Docker, Kafka

- Developed a scalable, real-time chat platform leveraging Socket.IO and Express.js to enable high-throughput message broadcasting and group communication, supporting 120+ concurrent users with sub-1ms message latency.
- Architected robust, secure RESTful APIs for user authentication via stateless JWTs, enabling protected routes and efficient session management.
- Engineered optimized, concurrent file upload mechanisms (95ms avg. response) using Multer for stream-based processing and managed distributed asset metadata in MongoDB with tailored indexing for data consistency.

RookieDB: Database Implementation | Java, SQL

- Implemented a SQL database engine with multi-granularity locking and ARIES recovery, ensuring data integrity, rapid recovery from failures, and consistent query results.
- Optimized query performance via B+ tree indices and join optimization algorithms enabling faster data retrieval for complex queries.

End-to-End Encrypted File Sharing | Go, Argon2, RSA, HMAC

- Developed a secure file-sharing solution to ensure data security on an untrusted server, maintaining confidentiality, integrity, and update/revocation controls.
- Optimized for large file scalability, achieving 99% bandwidth compliance between system and expected use through chunking large files, efficient stream processing, and asynchronous operations.
- Designed 4-layer cryptographic architecture with HMAC integrity verification, supporting multi-user collaboration with granular access control and efficient tree-based permission propagation.

Gitlet: Mini Version Control System | Java, Git

- Designed a local version control system to track file changes and version history for restoration and rollbacks.
- Applied SHA-1 hashing, tree-based data structures, and serialization to maintain integrity, history, and persistency.